

Brights

Always learning.

Fall26UDIGraduateDeveloperNO





PROGRAM OVERVIEW

C#.NET

MODULE 1 (4 DAYS)

GitHub & Docker + Relational Databases with PostgreSQL

- Select, focusing on Sub-query & JOINDDL & SQL insert/update/delete
- Proper use of Primary & Foreign Keys
- SQL transactions & ACID
- ER modelling
- DB best practices

MODULE 2 (1 DAY)

C# Intermediate

- Thorough C# and .NET knowledge
- Design guidelines (SOLID)
- Design patterns (DI and more)
- Unit testing
- ADO.NET
- Contributing to existing code

MODULE 3 (4 DAYS)

C# Database Integration

- Code to query data sources
- Entity Framework Core (ORM)
- Code-first setup & migrations
- DB integration testing
- SQL injection security threat
- Diagnose code-to-DB performance

MODULE 4 (5 DAYS)

C# Web API

- Develop ASP.NET Core Web API solution
- Authorization /authentication (with Oauth 2.0)
- Asynchronous processing with message queues
- API integration testing (using TestServer)
- Setting up pipelines for build /test / deployment

MODULE 5 (5 DAYS)

Mastering Self-leadership

- Self leadership
- Communication skills
- Critical thinking

MODULE 6 (4 DAYS)

TypeScript & React

- TypeScript
- React (JSX) syntax
- React Router
- Redux (RTK)
- Building UI with React
- Unit testing (React Testing Library)

MODULE 7 (3 DAYS)

Azure cloud

- Deploying to cloud solutions (Azure)

MODULE 8 (10 DAYS)

Final Project

- Tailor -made project

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Learning outcomes



LEARNING OUTCOMES

Module 3

Relational Databases with PostgreSQL

Days: 4

Teacher: Tomas Sandes

The Participant . . .

Knowledge

- ...understands the relational model, including proper use of keys.
- ...is familiar with ER modelling syntax.
- ...has good knowledge of advanced SQL queries, utilizing JOINS and sub-queries.
- ...understands stored procedures and their uses.
- ...has knowledge of indexes.
- ...understands the ACID principles, with focus on how to use SQL transactions.
- ...knows how to do user administration.

Skills

- ...can use an ER model to understand and navigate a database project and can build ER models for existing projects.
- ...has experience with PostgreSQL databases and relevant tools.
- ...can write efficient SQL statements for handling CRUD operations.
- ...can use DDL to create, update and delete databases and tables.
- ...has experience with deploying databases to cloud environments.

General competence

- ...is familiar with best practices when working on database projects.
- ...understands how to develop, test and deploy database solutions through proper use of dev, test, and prod environments.



LEARNING OUTCOMES

Module 4

C# intermediate

Days: 5

Teacher: Tomas Sandes

The Participant . . .

Knowledge

- ...has a thorough understanding of C# and .NET syntax, including several features from the latest versions.
- ...is familiar with industry-approved practices for code structuring, like design guidelines ...(SOLID) and patterns (Dependency Injection and others).
- ...understands version control systems (Git) and their terminology, like repository, branch, clone, merge, and rebase.
- ...understands how to write good test setups for software projects.

Skills

- ...has experience working on existing C# code bases through version control systems (Git), using commits and pull-push commands, and for solving bugs with the help of history/changes and the like.
- ...is familiar with best practices for testing, both for development and production code.
- ...can properly move production ready code from development to production environments.

General competence

- ...understands how to contribute to an existing code base through good code structure, test coverage, version control and handling of dev, test, and prod environments.



LEARNING OUTCOMES

Module 5

C# & Database Integration

Days: 5

Teacher: Tomas Sandes

The Participant . . .

Knowledge

- ...is familiar with C# basics for querying data sources, like LINQ method syntax.
- ...has a good understanding of the SQL injection security threat, and how to prevent it.
- ...knows how to properly inject a connection string and similar, sensitive data.
- ...knows how an ORM couples classes and entities, and how it translates LINQ to SQL statements.
- ...understands how to diagnose performance issues for code-to-DB communication.
- ...understands async and await regarding DB statement execution.

Skills

- ...is familiar with low level database communication with a DB driver, like ADO.NET.
- ...has experience with DB development through an ORM, like Entity Framework Core, including how to utilize migrations in code-first setups.
- ...can efficiently execute LINQ SQL statements through C# code.
- ...has experience in profiling code-to-DB communication, optimizing and fixing common pitfalls where necessary.

General competence

- ...can write efficient and robust code for communication with an SQL database.



LEARNING OUTCOMES

Module 6

C# Web API

The Participant . . .

Knowledge

- ...knows how to use ASP.NET Core Web API, and has knowledge of the other possible setups for ASP.NET Core (Minimal API, MVC, Razor pages and Blazor).
- ...is familiar with ASP.NET (with OAuth 2.0) for authorization/authentication.
- ...understands asynchronous processing, using message queues.
- ...knows how to use Postman or another API platform to test API functionality.
- ...understands how to properly take ASP.NET features from a development environment and deploy them to a production environment.

Skills

- ...has experience developing ASP.NET Core Web API solutions.
- ...has experience integrating with 3rd-party APIs.
- ...can use Docker and Docker compose for local development.
- ...has experience with ASP.NET Core integration tests, using TestServer or the like.

General competence

- ...can contribute efficiently in a full stack team for a live solution, utilizing dev, test and prod environments to their fullest.

Days: 4

Teacher: Bjørn



LEARNING OUTCOMES

Module 2

Vekstverkstedet

The Participant . . .

Knowledge

- ...has knowledge of the difference between a fixed and a growth mindset in a professional setting.
- ...knows how to identify personal triggers that inhibit growth and learning.
- ...understands the principles of active listening and qualified questioning.
- ...knows of various systems for knowledge sharing, such as pair programming and professional interest groups.
- ...understands the process of transforming critical feedback into constructive development goals.

Skills

- ...is familiar with mapping personal triggers and creating a measurable 30-day development plan.
- ...has experience using active listening and feedback techniques in professional communication.
- ...is familiar with turning criticism into actionable learning points through role-play or practical exercises.
- ...has experience defining a concrete plan for knowledge sharing within a team.
- ...is familiar with preparing and delivering a lightning talk or short presentation to share expertise.

General competence

- ..can contribute to a proactive learning culture by applying growth mindset principles and participating in structured knowledge-sharing rituals.

Days: 1

Teacher: Magnus Skaalsveen



LEARNING OUTCOMES

Module 1

TypeScript & React

The Participant . . .

Knowledge

- ...has knowledge of TypeScript syntax.
- ...has knowledge of React (JSX) syntax.
- ...knows how to use client-side routing (i.e using react-router).
- ...knows of alternatives to React (i.e., Vue, Angular, etc.).
- ...knows how to write unit tests for components and functions (react-testing-library, Jest).

Skills

- ...is familiar with contributing on a frontend application stack using TypeScript.
- ...has experience building user interface (UI) components using a frontend library (i.e., React).
- ...is familiar with using Storybook to develop and document UI components.
- ...has experience managing communication between the front-end and web APIs using HTTP and fetching approaches (i.e., using fetch and react-query).
- ...is familiar with advanced state management approaches (redux).
- ...is familiar with authentication using JSON web tokens (i.e Auth0 or OAuth 2.0).
- ...is familiar with deployment and continuous integration of a frontend stack.

General competence

- ...can contribute into a frontend application stack, providing features, bug fixes and tests.

Days: 7

Teacher: Magnus Skaalsveen



LEARNING OUTCOMES

Module 7

Azure Cloud

The Participant ...

Knowledge

- ...has insight in key Azure features and services, such as compute, storage, databases and network
- ...knows how to use Azure services securely
- ...understands the use of infrastructure as code

Skills

- ...can deploy Azure services
- ...can setup pipelines for builds, tests, and deployment using Azure DevOps
-can integrate applications with Azure platform services...can deploy to cloud solutions, like Azure App Service or Azure Kubernetes Service (AKS)

General competence

- ...understands the different service models of cloud providers...understands the responsibility distribution using cloud services...has insight in Azure features and services

Days: 3

Teacher: Bjørn



LEARNING OUTCOMES

Module 8

Final Project

The Participant ...

Knowledge

- .

Skills

-

General competence

-

Days: 10

Teacher: